

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
url = "https://raw.githubusercontent.com/nunnarilabs/ml/master/house/kc_house_data.csv"
df = pd.read_csv(url)
```

```
print(df.head())
print(df.info())
```

	id	date	price	bedrooms	bathrooms	sqft_living	\
0	7129300520	20141013T000000	221900.0	3	1.00	1180	
1	6414100192	20141209T000000	538000.0	3	2.25	2570	
2	5631500400	20150225T000000	180000.0	2	1.00	770	
3	2487200875	20141209T000000	604000.0	4	3.00	1960	
4	1954400510	20150218T000000	510000.0	3	2.00	1680	

	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_basement	\
0	5650	1.0	0	0	...	7	1180	0	
1	7242	2.0	0	0	...	7	2170	400	
2	10000	1.0	0	0	...	6	770	0	
3	5000	1.0	0	0	...	7	1050	910	
4	8080	1.0	0	0	...	8	1680	0	

	yr_built	yr_renovated	zipcode	lat	long	sqft_living15	\
0	1955	0	98178	47.5112	-122.257	1340	
1	1951	1991	98125	47.7210	-122.319	1690	
2	1933	0	98028	47.7379	-122.233	2720	
3	1965	0	98136	47.5208	-122.393	1360	
4	1987	0	98074	47.6168	-122.045	1800	

	sqft_lot15
0	5650
1	7639
2	8062
3	5000
4	7503

```
[5 rows x 21 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21613 entries, 0 to 21612
Data columns (total 21 columns):
```

#	Column	Non-Null Count	Dtype
0	id	21613 non-null	int64
1	date	21613 non-null	object
2	price	21613 non-null	float64
3	bedrooms	21613 non-null	int64
4	bathrooms	21613 non-null	float64
5	sqft_living	21613 non-null	int64
6	sqft_lot	21613 non-null	int64
7	floors	21613 non-null	float64
8	waterfront	21613 non-null	int64
9	view	21613 non-null	int64
10	condition	21613 non-null	int64
11	grade	21613 non-null	int64
12	sqft_above	21613 non-null	int64
13	sqft_basement	21613 non-null	int64
14	yr_built	21613 non-null	int64
15	yr_renovated	21613 non-null	int64
16	zipcode	21613 non-null	int64
17	lat	21613 non-null	float64

```
18 long          21613 non-null float64
19 sqft_living15  21613 non-null int64
20 sqft_lot15     21613 non-null int64
dtypes: float64(5), int64(15), object(1)
memory usage: 3.5+ MB
None
```

```
print("\nMissing Values:\n", df.isnull().sum())
```

```
Missing Values:
id          0
date        0
price       0
bedrooms    0
bathrooms   0
sqft_living  0
sqft_lot     0
floors       0
waterfront  0
view         0
condition    0
grade        0
sqft_above   0
sqft_basement 0
yr_built     0
yr_renovated  0
zipcode      0
lat          0
long         0
sqft_living15 0
sqft_lot15    0
dtype: int64
```

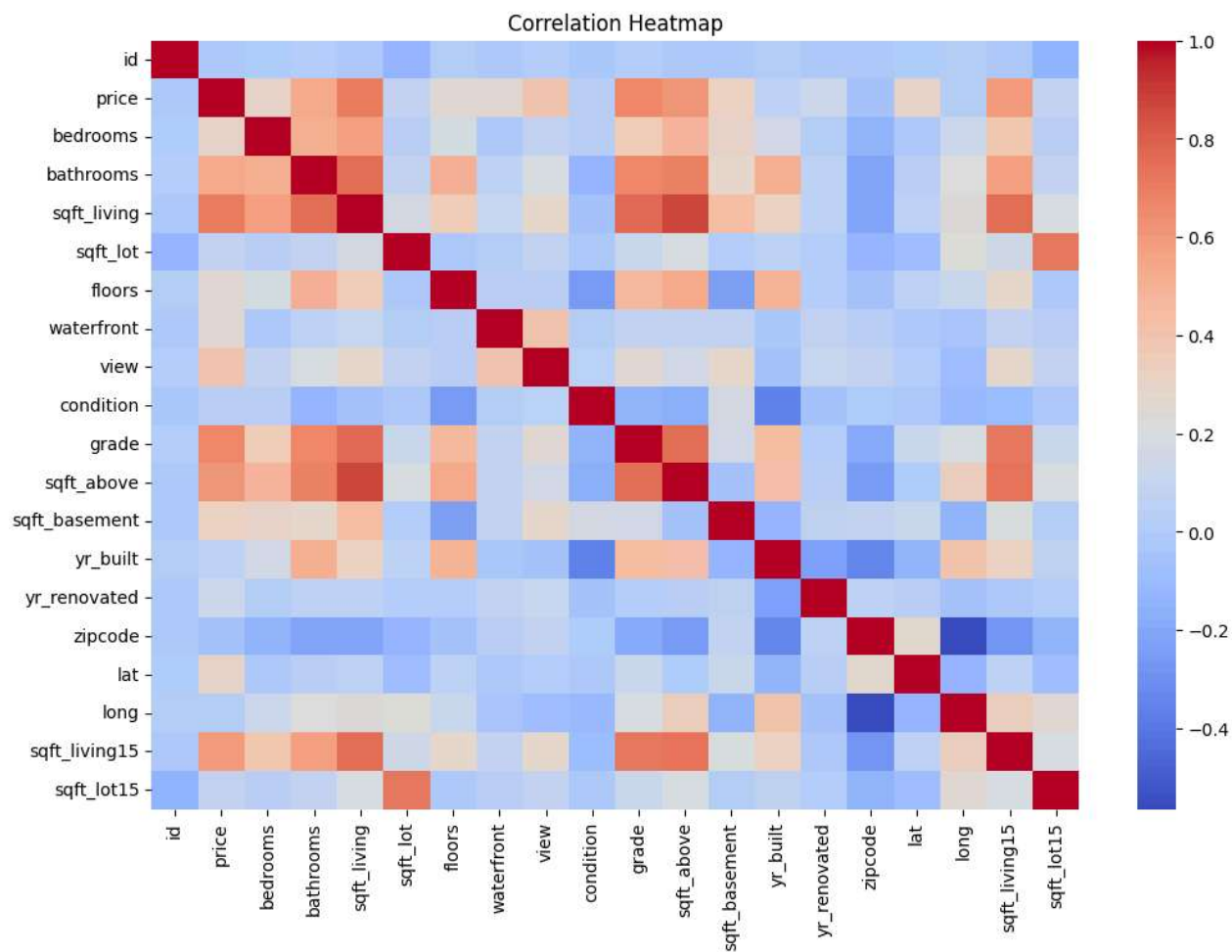
```
df = df.dropna()
```

```
corr_matrix = df.corr(numeric_only=True)
print("\nCorrelation Matrix:\n", corr_matrix)
```

condition	-0.158214	0.174105	-0.361417	-0.060618	0.003026
grade	0.755923	0.168392	0.446963	0.014414	-0.184862
sqft_above	1.000000	-0.051943	0.423898	0.023285	-0.261190
sqft_basement	-0.051943	1.000000	-0.133124	0.071323	0.074845
yr_built	0.423898	-0.133124	1.000000	-0.224874	-0.346869
yr_renovated	0.023285	0.071323	-0.224874	1.000000	0.064357
zipcode	-0.261190	0.074845	-0.346869	0.064357	1.000000
lat	-0.000816	0.110538	-0.148122	0.029398	0.267048
long	0.343803	-0.144765	0.409356	-0.068372	-0.564072
sqft_living15	0.731870	0.200355	0.326229	-0.002673	-0.279033
sqft_lot15	0.194050	0.017276	0.070958	0.007854	-0.147221

	lat	long	sqft_living15	sqft_lot15
id	-0.001891	0.020799	-0.002901	-0.138798
price	0.307003	0.021626	0.585379	0.082447
bedrooms	-0.008931	0.129473	0.391638	0.029244
bathrooms	0.024573	0.223042	0.568634	0.087175
sqft_living	0.052529	0.240223	0.756420	0.183286
sqft_lot	-0.085683	0.229521	0.144608	0.718557
floors	0.049614	0.125419	0.279885	-0.011269
waterfront	-0.014274	-0.041910	0.086463	0.030703
view	0.006157	-0.078400	0.280439	0.072575
condition	-0.014941	-0.106500	-0.092824	-0.003406
grade	0.114084	0.198372	0.713202	0.119248
sqft_above	-0.000816	0.343803	0.731870	0.194050
sqft_basement	0.110538	-0.144765	0.200355	0.017276
yr_built	-0.148122	0.409356	0.326229	0.070958
yr_renovated	0.029398	-0.068372	-0.002673	0.007854
zipcode	0.267048	-0.564072	-0.279033	-0.147221
lat	1.000000	-0.135512	0.048858	-0.086419
long	-0.135512	1.000000	0.334605	0.254451
sqft_living15	0.048858	0.334605	1.000000	0.183192
sqft_lot15	-0.086419	0.254451	0.183192	1.000000

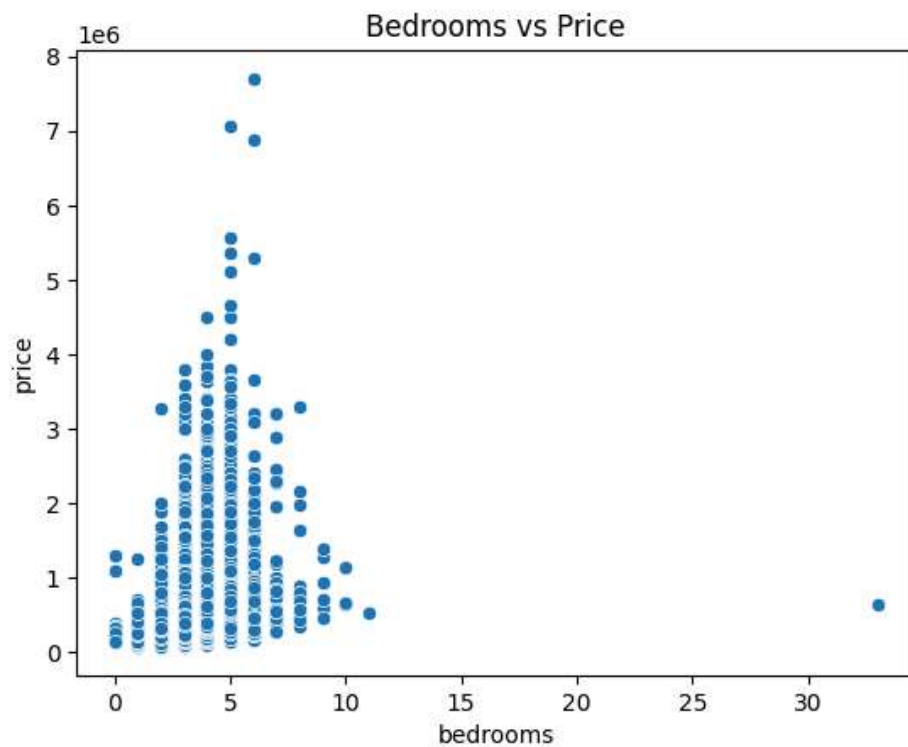
```
plt.figure(figsize=(12,8))
sns.heatmap(corr_matrix, annot=False, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



```
plt.figure()
plt.scatter(df['sqft_living'], df['price'])
plt.xlabel("Sqft Living")
plt.ylabel("Price")
plt.title("Sqft Living vs Price")
plt.show()
```



```
plt.figure()
sns.scatterplot(x='bedrooms', y='price', data=df)
plt.title("Bedrooms vs Price")
plt.show()
```



```
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)
```

```
print(df.isnull().sum())
```

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

```
df['Age'].fillna(df['Age'].mean(), inplace=True)
```

/tmp/ipykernel_1709/1226013979.py:1: FutureWarning: A value is trying to be set on a copy of a Data
The behavior will change in pandas 3.0. This inplace method will never work because the intermedia

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value},

```
df['Age'].fillna(df['Age'].mean(), inplace=True)
```

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

/tmp/ipykernel_1709/411043493.py:1: FutureWarning: A value is trying to be set on a copy of a Data
The behavior will change in pandas 3.0. This inplace method will never work because the intermedia

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value},

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

```
print(df.isnull().sum())
```

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         0
dtype: int64
```

```
df.drop(['Cabin', 'Ticket', 'PassengerId'], axis=1, inplace=True)
```

```
print(df.columns)
```

```
Index(['Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare',
      'Embarked'],
      dtype='object')
```

```
df.rename(columns={
    'Pclass': 'Passenger_Class',
    'SibSp': 'Siblings_Spouses',
    'Parch': 'Parents_Children'
}, inplace=True)
```

```
print(df.columns)
```

```
Index(['Survived', 'Passenger_Class', 'Name', 'Sex', 'Age', 'Siblings_Spouses',
      'Parents_Children', 'Fare', 'Embarked'],
      dtype='object')
```

```
df['Age'] = df['Age'].astype(int)
df['Survived'] = df['Survived'].astype(bool)
df['Passenger_Class'] = df['Passenger_Class'].astype('category')
```

```
print(df.dtypes)
```

```
Survived          bool
Passenger_Class   category
Name              object
Sex               object
Age               int64
Siblings_Spouses  int64
Parents_Children  int64
Fare              float64
Embarked          object
dtype: object
```

```
print(df['Sex'].value_counts())
print(df['Embarked'].value_counts())
print(df['Passenger_Class'].value_counts(normalize=True))
```

```
Sex
male      577
female    314
Name: count, dtype: int64
Embarked
S      646
C     168
Q      77
Name: count, dtype: int64
Passenger_Class
3    0.551066
1    0.242424
2    0.206510
Name: proportion, dtype: float64
```

```
print(df.groupby('Sex')['Fare'].mean())
print(df.groupby('Passenger_Class')['Age'].mean())
```

```
Sex
female    44.479818
male      25.523893
Name: Fare, dtype: float64
Passenger_Class
1    36.944444
```

```

2    29.804348
3    26.191446
Name: Age, dtype: float64
/tmp/ipykernel_1709/2471021283.py:2: FutureWarning: The default of observed=False is deprecated an
print(df.groupby('Passenger_Class')['Age'].mean())

```

```

corr = df.corr(numeric_only=True)
print(corr)

```

	Survived	Age	Siblings_Spouses	Parents_Children	\
Survived	1.000000	-0.067809	-0.035322	0.081629	
Age	-0.067809	1.000000	-0.232743	-0.176744	
Siblings_Spouses	-0.035322	-0.232743	1.000000	0.414838	
Parents_Children	0.081629	-0.176744	0.414838	1.000000	
Fare	0.257307	0.093856	0.159651	0.216225	

	Fare
Survived	0.257307
Age	0.093856
Siblings_Spouses	0.159651
Parents_Children	0.216225
Fare	1.000000

```
print(corr['Fare'].sort_values(ascending=False))
```

Fare	1.000000
Survived	0.257307
Parents_Children	0.216225
Siblings_Spouses	0.159651
Age	0.093856

Name: Fare, dtype: float64

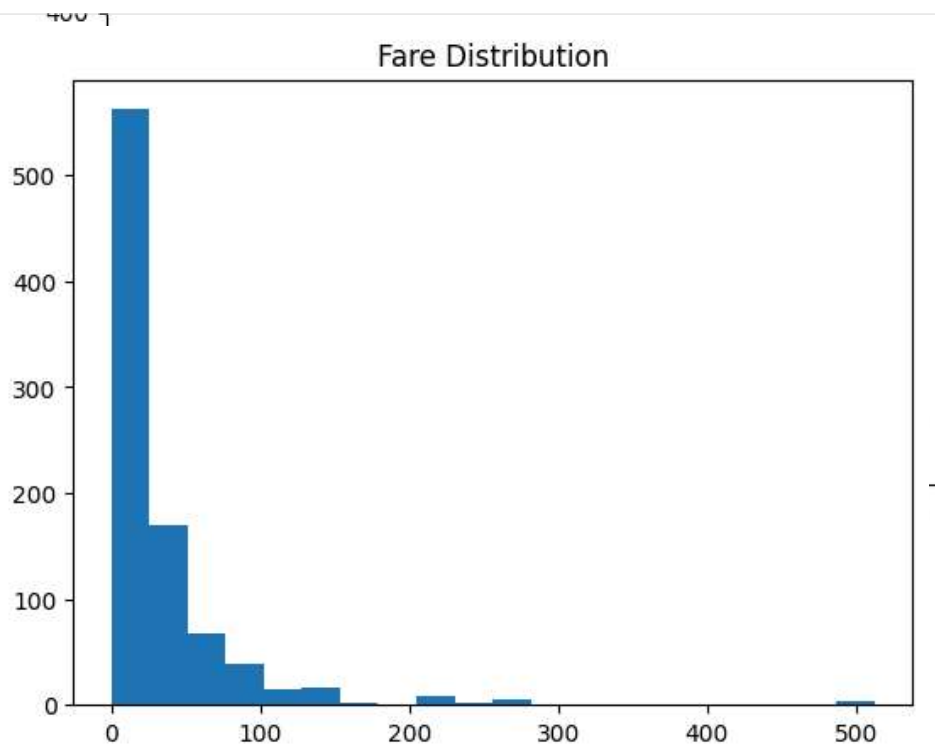
```

plt.scatter(df['Age'], df['Fare'])
plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Age vs Fare")
plt.show()

```


Age vs Fare

```
plt.hist(df['Fare'], bins=20)  
plt.title("Fare Distribution")  
plt.show()
```



```
plt.figure(figsize=(10,6))  
sns.heatmap(corr, annot=True, cmap='coolwarm')  
plt.show()
```